

SUPERVISED MACHINE LEARNING: CLASSIFICATION

MODULE 6:

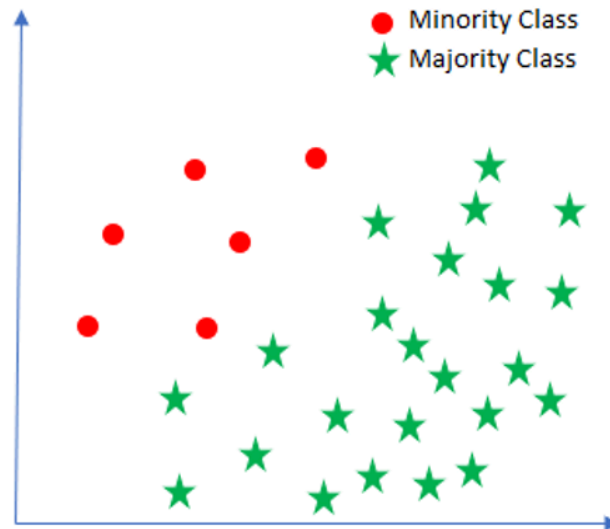
MODELING UNBALANCED CLASSES

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Problem Overview

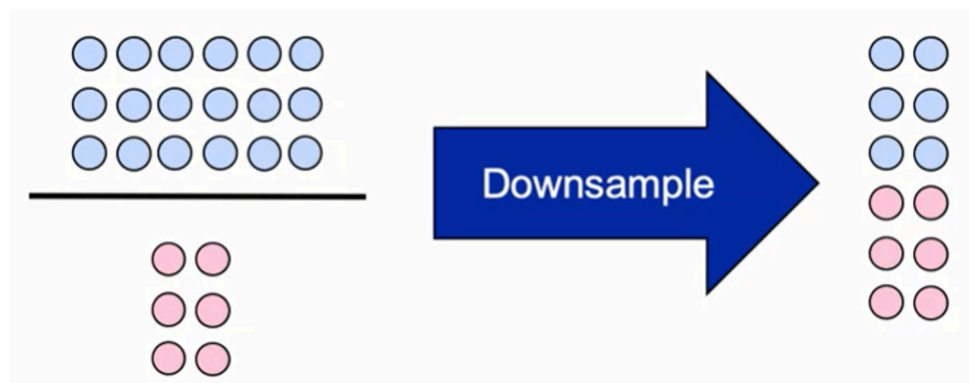
Imbalanced Data



- Most classification algorithms aim to maximize accuracy (độ chính xác).
- When one class dominates, the model tends to predict that class → poor recall (độ bao phủ) on minority class.
- Goal: balance the dataset before training.

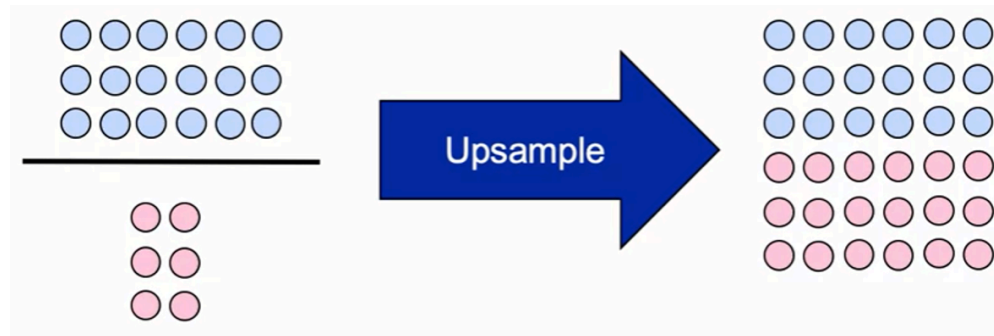
Sampling Strategies

- **Downsampling (Giảm mẫu)**



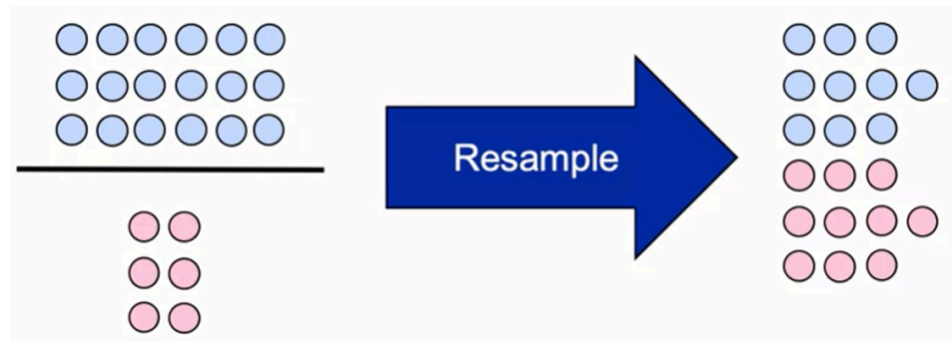
- Remove observations from **majority class**.
- Reduces data volume → may lose information.
- Increases **recall**, decreases **precision**.

- **Upsampling (Tăng mẫu)**



- Duplicate or synthetically generate observations for **minority class**.
- Keeps data volume but may cause **overfitting**.
- Increases **precision**, slightly lowers **recall**.

- **Resampling (Kết hợp)**

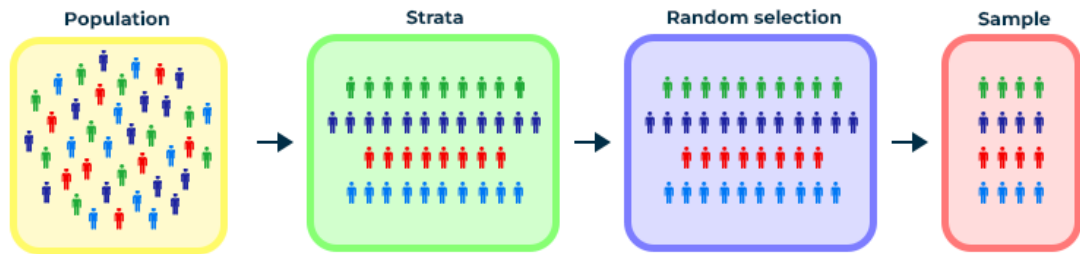


- Apply both: reduce majority class and duplicate minority class until balance achieved

```
# Split before sampling to avoid data leakage
X_train, X_test, y_train, y_test = train_test_split(X, y,
stratify=y)
```

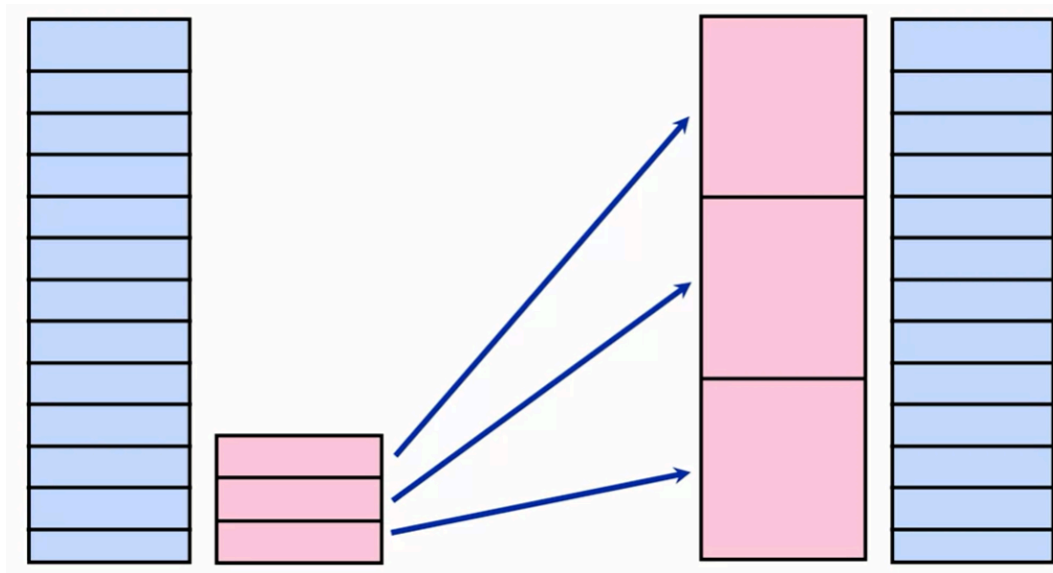
Key Methods for Resampling

- **Stratified Sampling (Lấy mẫu phân tầng)**



- Preserves class proportions across **train/test** datasets.
- Use parameter `stratify=y` in `train_test_split`.

- **Random Oversampling (Tăng mẫu ngẫu nhiên)**

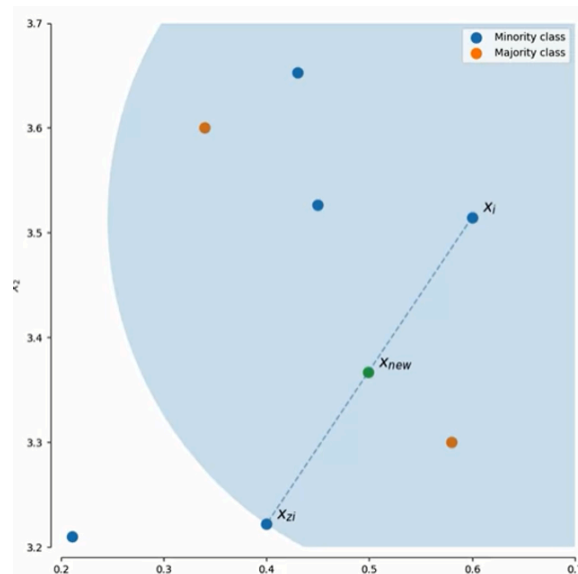


Oversampling

- Randomly duplicates minority samples with replacement.

```
from imblearn.over_sampling import RandomOverSampler
X_res, y_res = RandomOverSampler().fit_resample(X, y)
```

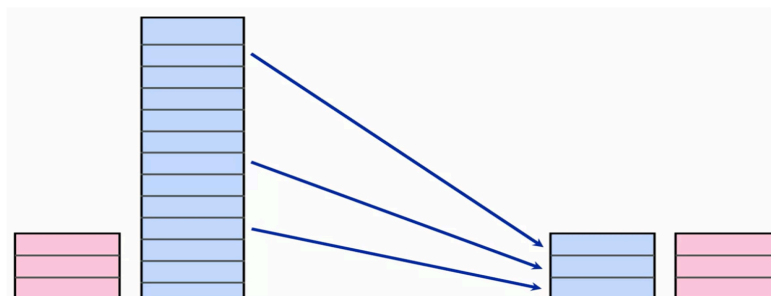
- **Synthetic Oversampling (Tăng mẫu tổng hợp)**



- SMOTE (Synthetic Minority Oversampling Technique):
 - Creates synthetic samples along lines between minority points and neighbors.
 - Variants: `Borderline-SMOTE`, `SVM-SMOTE`.
- ADASYN (Adaptive Synthetic Sampling):
 - Focuses on harder-to-learn regions (minority near decision boundary).

```
from imblearn.over_sampling import SMOTE, ADASYN
X_smote, y_smote = SMOTE().fit_resample(X, y)
X_ada, y_ada = ADASYN().fit_resample(X, y)
```

- **Undersampling (Giảm mẫu)**



- Removes majority samples to match minority class.
- Methods:
 - **NearMiss:** keeps majority samples closest to minority ones.
 - **Tomek Links:** removes overlapping majority–minority pairs.
 - **Edited Nearest Neighbors (ENN):** removes misclassified samples.

```
from imblearn.under_sampling import TomekLinks, NearMiss
X_tomek, y_tomek = TomekLinks().fit_resample(X, y)
X_near, y_near = NearMiss().fit_resample(X, y)
```

Advanced Techniques

- **Blagging (Balanced Bagging)**
 - Ensemble of bootstrapped samples where **each subset is balanced** via downsampling.
 - Reduces bias toward majority class while leveraging ensemble stability.

```
from imblearn.ensemble import BalancedBaggingClassifier
from sklearn.tree import DecisionTreeClassifier
bbc = BalancedBaggingClassifier(
    base_estimator=DecisionTreeClassifier(),
    n_estimators=50)
bbc.fit(X_train, y_train)
```

- **Class Weights (Trọng số lớp)**
 - Alternative to sampling.
 - Add parameter `class_weight='balanced'` in many `sklearn` models.

```
from sklearn.linear_model import LogisticRegression
model = LogisticRegression(class_weight='balanced')
model.fit(X_train, y_train)
```

Evaluation Metrics

Avoid **accuracy**, which is misleading. Use:

- **Recall (Độ bao phủ)**: captures minority class well.
- **Precision (Độ chính xác)**: measures prediction reliability.
- **F1-score**: harmonic mean of precision and recall.
- **AUC-ROC**: trade-off between true/false positives.
- **Cohen's Kappa**: measures agreement beyond chance.

```
from sklearn.metrics import classification_report,
roc_auc_score
print(classification_report(y_test, y_pred))
roc_auc_score(y_test, model.predict_proba(X_test)[: ,1])
```

Key Takeaways

Method	Description	Pros	Cons
Downsampling	Reduce majority samples	↑ Recall	↓ Precision, data loss
Upsampling	Duplicate minority samples	↑ Precision	Overfitting
SMOTE/ ADASYN	Synthetic generation	Better boundary coverage	Slower
NearMiss/ Tomek ENN	Undersampling variants	Clearer separation	Risk of data loss
Blagging	Balanced bagging ensemble	Parallel, stable	Computational cost